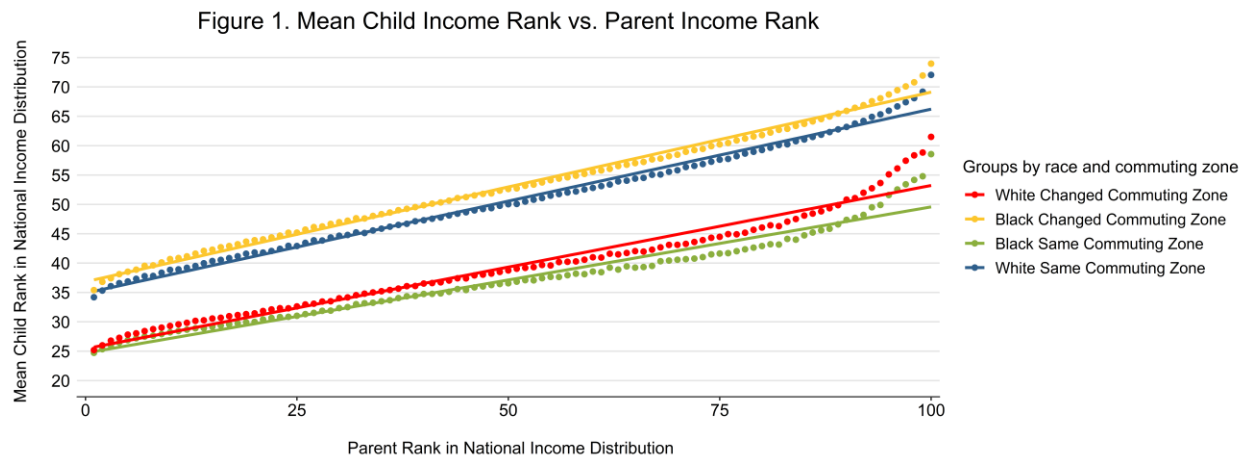


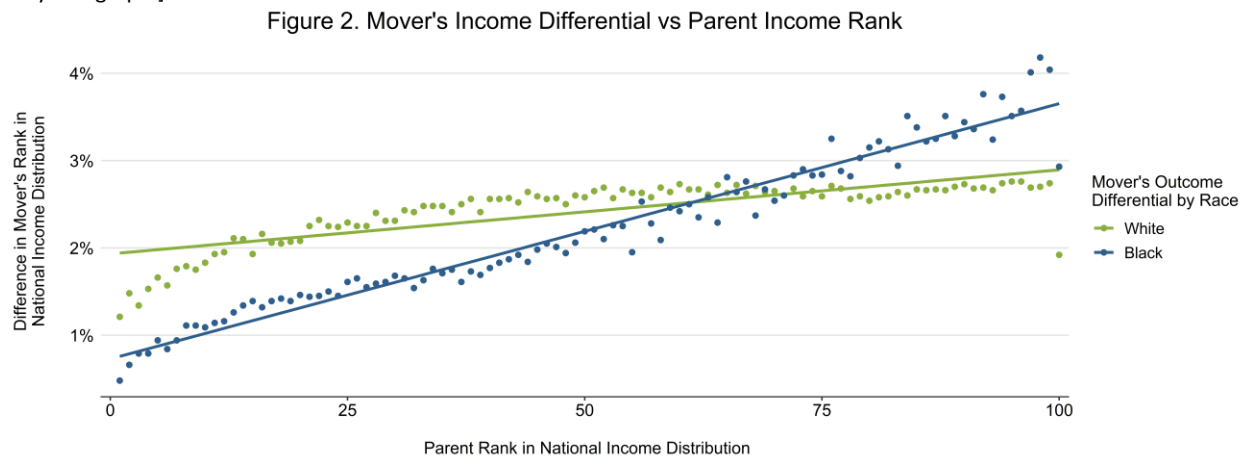
Question 1

In our lab we often represent income with a person's percentile rank in the national income distribution. The CSV file ***national_percentile_outcomes.csv*** lists the average household income percentile of children given their race and the income percentile of their parents. Percentile ranks are listed for all children as well as for children who stayed as adults in the same Commuting Zone (CZ) where they grew up. Plot separate series for each race. Show the slopes of all 4 series in the graph. [You should produce only one graph].

- a) Present a scatter plot showing the relationship between child and parent household income percentile ranks using parent income on the x-axis. Do so separately for all children and for children who stayed in the same CZ where they grew up. Plot separate series for each race. Show the slopes of all 4 series in the graph. [You should produce only one graph].



- b) Now compute the absolute difference in child household income percentile ranks between all children and children that stay in their CZ at each parent income rank. Do so for black and for white children separately. Present a new scatter plot showing the relationship of the two differences in ranks with parental income. Include fitted lines for both series. [You should produce only one graph.]



Question 2

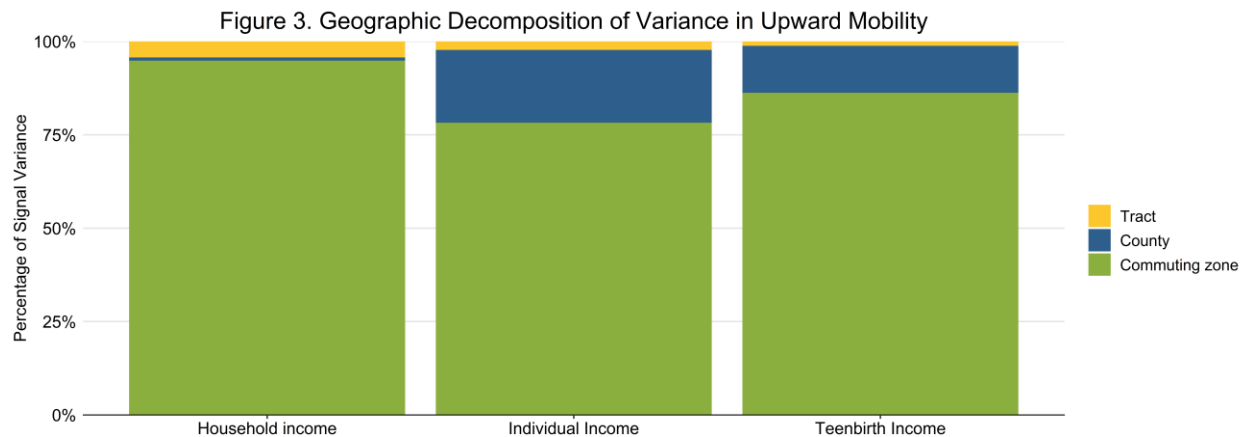
Economic mobility varies across place. The CSV file ***tract_outcomes.csv*** lists neighborhood- specific measures of economic mobility. In particular, it contains several different outcomes for children whose parents were in the 25th percentile of the parent income distribution for children who grew up in a particular Census tract. Each outcome is presented for pooled race and gender as well as separately by race and by gender. You can read ***tract_outcomes_codebook.pdf*** for a more detailed description of all variables. The CSV file ***cz_county_crosswalk.csv*** contains a crosswalk from county to Commuting Zone.

We want to explore how much of the variation in the outcomes for females is explained at different levels of geography.

- a) Present a bar graph showing the share of total variance of three outcomes explained by tract, county and commuting zone fixed effects for females with parents at the 25th percentile of the distribution: individual income percentile ranks, household income percentile ranks, and teen birth rates.

Your plot should have three stacked bars, one for each outcome. Construct each of the stacked bars by first plotting a bar showing the percent of variance explained across tracts (always equal to 100% because you are working with tract level data), then overlay a bar showing the percent of variance explained across counties, and finally overlay a bar showing the percent of variance explained across commuting zones. Present y-axis labels as percentages.

[Help: You can compute the share of total variance of an outcome explained across a geographic level as the adjusted R squared from a regression of the outcome on that geographic level's fixed effects. Don't forget to use the right population weights!]



b) Based on the graph you constructed in (a), which of the outcomes presented seems to have more local variation? In one sentence explain why you conclude that from the graph. For the outcome that you chose as an answer, list the percentage of total variance explained at the county level (present it as a percentage rounded to nearest integer).

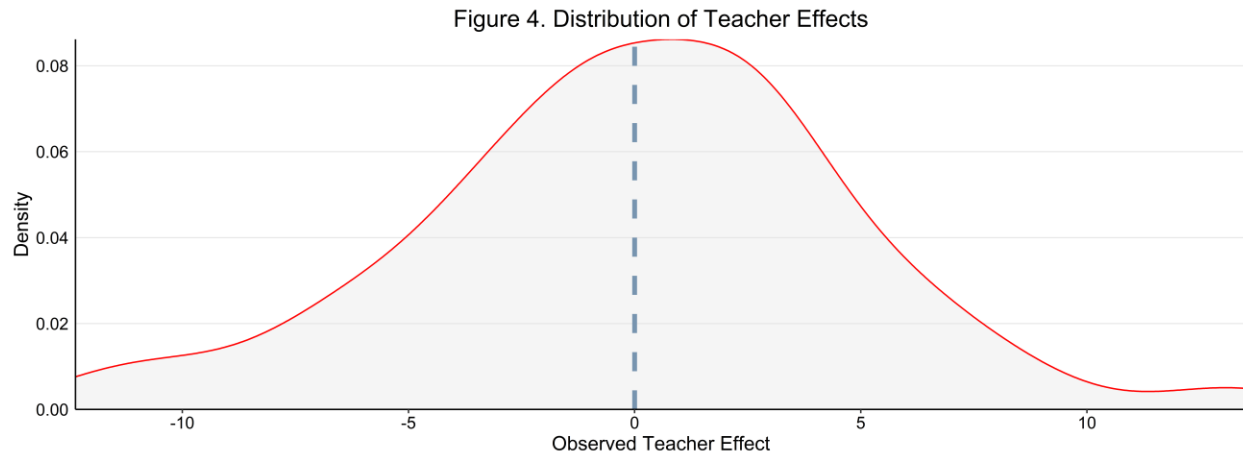
Individual income. Has the largest portion of variance explained outside of commuting zone. (County = 19%)

Question 3

Previous research at Opportunity Insights has shown that teachers have large effects on student outcomes. In particular, some teachers have much more positive impacts on their students than others.

Imagine that School District A randomly assigns teachers and students to fifth-grade classes and measures the average standardized test score of students in each teacher's class. School District A then calculates the class average of each teacher minus the mean for the school district as a whole. School District A refers to this measure as the effect of the teacher on scores. Assume that School District A has effectively randomized students across teachers, such that each teacher has similar students on average.

a) The CSV file **teacher_effects.csv** gives School District A's estimates of each teacher's estimated effect on fifth-grade math scores. Plot the distribution of estimated teacher effects. Include lines to indicate the mean on the graph, and calculate the standard deviation.



Standard Deviation: 4.817082

b) We will refer to the true effect of teacher tt on average scores as ss_{tt} , and the measured effect of teacher tt on average scores as $\bar{s}s_{tt}$. How would the distribution of ss_{tt} differ from the distribution of $\bar{s}s_{tt}$? Make clear any assumptions you are making when you answer this question. Provide a short intuitive answer and a brief mathematical proof/ reasoning.

The distribution of ss_{tt} may differ from distribution of the average scores $\bar{s}s_{tt}$ because the average score is calculated using all of District A's students. Differences in grade-level curriculums may mean some grade-levels are harder than others. Thus, comparing 5th grader teacher's scores to an average which includes teacher's scores from other grades may introduce bias (e.g. If the 5th grade is the hardest grade then a teachers scores may be lower than the district average. If the 5th grade is easier, then their measured effect may appear to higher than average.) Assuming that the test differs across grades, this would also impact the distribution of scores.

Additionally, if sunlight positively affects student outcomes, then teachers in south-facing rooms will have above average effects on students (measured by their score). This upward bias is attributed to the teacher even though the physical provides them with an advantage. Because teacher scores are calculated by subtracting the average score within sample deviations can introduce omitted variable bias (e.g. newer schools may bias effectiveness scores upwards because they improve morale; thus the facilities provide and advantage because they are better than the district's average). Randomly assigning teachers and students to classes does not eliminate this bias. Including school and time fixed effects would be a better method of evaluation, thus you could control for location by randomizing school assignments across years to eliminate these effects.

c) School District B is impressed with the study and hires the ten teachers with the highest estimated effects on scores. A year later, School District B measures the effectiveness of each teacher, and finds that the teachers they hired from School District A are much less effective than they had expected. The superintendent of School District B asks you to explain what's happened. Assume that School Districts A and B are similar in every respect, and that teachers have been placed into schools in School District B that are similar to the schools in which they taught in School District A. Assume also that the school districts have used the same methodology to estimate the effect of each teacher on scores. Provide a short intuitive answer.

The teachers hired from District A may not be effective because only one year's worth of data (scores) is being considered. This approach is analogous to picking stocks after looking at the previous year's performance. Sometimes, you are bound to pick teachers whose performance are anomalies. Even with random assignment, someone has to hit the lottery. As your advisor, I recommend hiring teachers whose scores are consistently above average. I also would caution basing hiring decisions off one year's worth of data. Some teachers you hired may have had south-facing rooms in their previous schools boosting their effectiveness. Even if that were not the case, there are other explanations for a loss of effectiveness. The best way to prevent this from happening is to look at a longer time horizon. To underscore my point, think of it intuitively. Hiring a teacher with one year's worth of performance is inherently more prone to mistakes than hiring a teacher with 20 years of above-average performance. Even the best teachers have off years, however, over time the cream will always rise to the top!